

D610

Data Engineering Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Analysts at a regional hospital network spend three days every month downloading CSV files from 6 systems and merging them in spreadsheets. The network recently won a quality award. Which solution is data engineering best suited to deliver?

- A. A better spreadsheet template for the monthly merge
- B. Automated pipelines that ingest the 6 systems into a central store on a schedule
- C. A machine learning model that predicts readmissions
- D. A report celebrating the quality award

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A capstone learner designs a pipeline for a fictional bike-share company with 3,000 bikes in 2 cities. Which layer order is correct?

- A. Sources → processing → ingestion → dashboards
- B. Sources → ingestion → raw storage → processing → curated serving store → dashboards
- C. Sources → ingestion → dashboards
- D. A different order for each city

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

In a data architecture, object storage that holds raw, unmodified copies of source files belongs to which layer?

- A. The processing layer
- B. The source system
- C. The raw (landing) storage layer
- D. The serving layer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A capstone team built its cloud resources by clicking through the web console, and now a teammate cannot reproduce the environment. The team coordinates in a shared chat channel. What practice fixes this?

- A. Write the click steps in a wiki page.
- B. Take screenshots of each console screen.
- C. Post updates more often in the chat channel.
- D. Define the infrastructure as code in version control.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A pipeline stores 2 TB (2,000 GB) in object storage at \$0.023 per GB-month and runs a transform cluster 3 hours a day at \$4 per hour for 30 days. The team’s logo is blue. What is the monthly cost?

- A. \$58
- B. \$2,926
- C. \$360
- D. \$406

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A project budget allows \$500 a month. Storage costs \$100 a month, and compute costs \$5 an hour. How many compute hours fit in the budget each month?

- A. 100 hours
- B. 120 hours
- C. 80 hours
- D. 20 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A bike-share analytics model records each ride with start station, end station, rider type, start date, and duration in minutes. The company's bikes are painted green. What is the fact table's grain, and what is the measure?

- A. One row per day with total duration.
- B. One row per bike color.
- C. One row per ride; duration is the measure, and stations, rider type, and date are dimensions.
- D. One row per station; duration is a dimension.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A fact table has one row per product per store per day, with units_sold as the measure. Which question can it NOT answer?

- A. Which customer bought the most units?
- B. Units by product by day
- C. Total units per store last month
- D. Average daily units per store

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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9.

A food-delivery firm operating in 15 cities has a pipeline that reports “success” every night, but yesterday’s load contained 0 rows because the source API returned nothing. What monitoring would have caught it?

- A. Row-count and freshness checks with alert thresholds
- B. Adding more cities to the feed
- C. Job success status alone
- D. CPU and memory graphs

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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10. [TRANSFER — reversed stem / different variable]

An SLA says dashboard data must be no more than 2 hours old. The last successful load finished at 6:00 a.m., and it is now 8:30 a.m. What is the status?

- A.** Breached by 30 minutes
- B.** Breached by 2.5 hours
- C.** Within the SLA
- D.** Unknown without row counts

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A capstone pipeline stores rider trip histories that reveal home-area travel patterns. All 5 team members have admin rights on every storage bucket. The project demo is on Friday. What is the best governance step?

- A.** Postpone the demo.
- B.** Keep everyone as admin to move faster.
- C.** Encrypt the buckets; encryption makes access control unnecessary.
- D.** Apply least-privilege roles, encrypt data, and mask or aggregate location fields not needed for analysis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

After building the pipeline, the learner presents to the bike-share company's operations director. The solution uses 14 cloud services. What should the presentation emphasize?

- A.** A walk-through of all 14 cloud services.
- B.** The outcome: hourly station data now drives bike rebalancing, with architecture detail as backup.
- C.** Only the architecture diagram, since outcomes never matter to operations.
- D.** A line-by-line code review.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

Analysts at a regional hospital network spend three days every month downloading CSV files from 6 systems and merging them in spreadsheets. ~~The network recently won a quality award.~~ Which solution is data engineering best suited to deliver?

Target word: fit of data engineering **Decides:** The pain is manual extraction and merging.

A	Dropped element	Keeps the manual work.
B	Correct	Removes the manual pulls and merges.
C	Adjacent category	Does not fix how data is gathered.
D	Camouflage	The award does not define the problem.

2. Answer B Concept 2

A capstone learner designs a pipeline for a fictional bike-share company ~~with 3,000 bikes in 2 cities.~~ Which layer order is correct?

Target word: order of layers **Decides:** Data must land before it is transformed and served.

A	Inverted operation	Data cannot be processed before it is ingested.
B	Correct	Each layer depends on the one before it.
C	Dropped element	Skips storage and transformation.
D	Camouflage	City count does not change the architecture.

3. Answer C Concept 2 · transfer

In a data architecture, object storage that holds raw, unmodified copies of source files belongs to which layer?

Target word: reverse: from component to layer **Decides:** Raw, unmodified copies.

A	Adjacent category	Transforms data; it does not store raw copies.
B	Confusion set	Sources create the data; the copy lives in the pipeline.
C	Correct	Keeps source data as received for reprocessing.
D	Inverted operation	Holds curated, query-ready data at the end of the flow.

4. Answer D Concept 3

A capstone team built its cloud resources by clicking through the web console, and now a teammate cannot reproduce the environment. ~~The team coordinates in a shared chat channel.~~ What practice fixes this?

Target word: reproducible environments **Decides:** Resources exist only as manual clicks.

A	Adjacent category	Still manual and drifts over time.
B	Dropped element	Cannot rebuild anything automatically.
C	Camouflage	Communication does not make infrastructure reproducible.
D	Correct	Anyone can rebuild the same environment from the code.

5. Answer D Concept 4

A pipeline stores 2 TB (2,000 GB) in object storage at \$0.023 per GB-month and runs a transform cluster 3 hours a day at \$4 per hour for 30 days. ~~The team's logo is blue.~~ What is the monthly cost?

Target word: total monthly cost **Decides:** Storage \$46 plus compute 90 hours × \$4.

A	Dropped element	Counts compute for one day only: \$46 + \$12.
B	Confusion set	Bills compute 24 hours a day: 720 × \$4 + \$46.
C	Dropped element	Leaves out storage.
D	Correct	2,000 × \$0.023 = \$46; 3 × 30 × \$4 = \$360; total \$406.

6. Answer C Concept 4 • transfer

A project budget allows \$500 a month. Storage costs \$100 a month, and compute costs \$5 an hour. How many compute hours fit in the budget each month?

Target word: solve for hours **Decides:** \$400 remains for compute.

A	Dropped element	Ignores the storage cost.
B	Inverted operation	Adds storage to the budget instead of subtracting.
C	Correct	$(\$500 - \$100) / \$5 = 80$.
D	Wrong stem number	Divides the storage cost by the hourly rate.

7. Answer C Concept 5

A bike-share analytics model records each ride with start station, end station, rider type, start date, and duration in minutes. ~~The company's bikes are painted green.~~ What is the fact table's grain, and what is the measure?

Target word: grain and measure **Decides:** The event being recorded is a ride.

A	Dropped element	Loses ride-level detail the model captures.
B	Camouflage	Color is not in the model and does not set the grain.
C	Correct	The ride is the event; duration is the numeric fact.
D	Inverted operation	Swaps roles: stations describe rides, duration is measured.

8. Answer A Concept 5 • transfer

A fact table has one row per product per store per day, with units_sold as the measure. Which question can it NOT answer?

Target word: reverse: limits of the grain **Decides:** No customer detail in the grain.

A	Correct	Customer is below the table's grain.
B	Adjacent category	That is the table's own grain.
C	Inverted operation	Answerable by summing daily rows.
D	Confusion set	Answerable by averaging store-day totals.

9. Answer A Concept 6

A food-delivery firm ~~operating in 15 cities~~ has a pipeline that reports “success” every night, but yesterday’s load contained 0 rows because the source API returned nothing. What monitoring would have caught it?

Target word: data-level monitoring **Decides:** The job succeeded but the data was empty.

A	Correct	Detect empty or stale loads regardless of job status.
B	Camouflage	City count does not detect empty loads.
C	Dropped element	That is what reported success.
D	Adjacent category	Show system health, not data content.

10. Answer A Concept 6 · transfer

An SLA says dashboard data must be no more than 2 hours old. The last successful load finished at 6:00 a.m., and it is now 8:30 a.m. What is the status?

Target word: freshness against SLA **Decides:** Data age is 2.5 hours.

A	Correct	2.5 hours old vs. a 2-hour limit.
B	Adjacent category	That is the data’s age, not the overage.
C	Inverted operation	2.5 hours exceeds 2 hours.
D	Confusion set	Freshness depends only on load time.

11. Answer D Concept 7

A capstone pipeline stores rider trip histories that reveal home-area travel patterns. All 5 team members have admin rights on every storage bucket. ~~The project demo is on Friday.~~ What is the best governance step?

Target word: protect sensitive data **Decides:** Sensitive data with broad admin access.

A	Camouflage	The schedule does not fix access control.
B	Inverted operation	Widens exposure of sensitive data.
C	Absolute language	Anyone with access still reads decrypted data.
D	Correct	Limits who can see what and reduces exposure.

12. Answer B Concept 8

After building the pipeline, the learner presents to the bike-share company’s operations director. ~~The solution uses 14 cloud services.~~ What should the presentation emphasize?

Target word: story for the stakeholder **Decides:** The audience runs operations, not infrastructure.

A	Camouflage	Service count does not matter to this audience.
B	Correct	Ties the build to the director’s decisions.
C	Absolute language	Outcomes are exactly what operations cares about.
D	Adjacent category	Suits engineers, not an operations director.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	When data engineering is the solution	Data engineering fits problems of moving, integrating, and serving data reliably, such as replacing manual extracts with automated pipelines into a central store.	Reaching for a predictive model when the pain is manual data handling.	People spending days pulling and merging files.
2	Layers of a data architecture	Sources → ingestion → raw (landing) storage → processing/transformation → curated serving store → analytics and dashboards.	Skipping the raw layer or transforming before ingestion.	A question about the order or role of a layer.
3	Infrastructure as code	Define cloud resources in version-controlled code so environments can be reproduced, reviewed, and rebuilt.	Relying on console clicks, screenshots, or wiki notes.	A teammate cannot recreate the environment.
4	Cloud cost estimation	Monthly cost = storage (GB × rate) + compute (hours actually run × hourly rate).	Forgetting one component or billing compute around the clock.	Storage size, rates, and daily run hours.
5	Fact table grain and dimensions	Set the grain first (what one fact row means); measures are numeric facts at that grain, and dimensions describe them. A fact table cannot answer questions below its grain.	Mixing grains or treating measures as dimensions.	“One row per...” or a question needing more detail than the table stores.
6	Data observability	Monitor the data itself (row volume, freshness, schema, null rates) against thresholds, not just job success.	Trusting a green job status or CPU graphs as proof the data is right.	A pipeline reports success but the data is wrong or stale.
7	Security and governance by design	Grant least privilege, encrypt in transit and at rest, minimize or mask sensitive fields, and track lineage in a catalog.	Giving everyone admin rights for speed, or thinking encryption replaces access control.	Sensitive data plus broad access.
8	Telling the story of the solution	Present the business outcome the pipeline enables to the decision maker; keep architecture detail as backup.	Leading with the list of services or the code.	A non-technical stakeholder audience for a technical build.

Course scope (public description): The Data Engineering Capstone has learners utilize the skills learned throughout the MDSA core courses and the data engineering courses to examine a problem where data engineering is a solution and to build a cloud-native infrastructure that allows for data processing. Learners are asked to implement their solutions and tell a story using the data. Course material introduces the project and reminds learners of relevant learning resources from previous courses that will prove helpful in completing the performance assessment.